Fractions: Simplifying

Simplifying Fractions

A fraction is said to be in **simplest form** or **lowest terms** when the numerator and denominator have no common factors other than 1. For example, the fraction $\frac{2}{5}$ is in simplest form because 2 and 5 have no common factors. However, the fraction $\frac{4}{6}$ is not in simplest form because both 4 and 6 have a common factor of 2. The process of writing a fraction in simplest form is called **simplifying** the fraction. We will consider two methods to simplify fractions. The first method uses the GCF and the second method uses prime factorizations.

Procedure 1 (**Simplifying Fractions Using the GCF**)**.** Divide the numerator and denominator by the GCF.

Example 1. Simplify the fraction $\frac{4}{12}$ using the GCF $= 4$.

Explanation. Divide the numerator and denominator by 4 and write the result.

$$\frac{4}{12} = \frac{4 \div 4}{12 \div 4}$$

$$= \frac{1}{3}$$

Done!

We can also simplify fractions by dividing the numerator and denominator by *any* common factor, however using the GCF involves just one step. If we didn't notice that the GCF was 4, but knew that both numbers were evenly divisible by 2, we would need two steps to simplify the fraction $\frac{4}{12}$. First, we divide the numerator and denominator by 2, then we divide by 2 again.

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$$\frac{4}{12} = \frac{4 \div 2}{12 \div 2}$$

$$= \frac{2}{6}$$

$$= \frac{2 \div 2}{6 \div 2}$$

$$= \frac{1}{3}$$

Problem 1 *Simplify the fraction* $\frac{24}{36}$ *using the GCF* $= 12$.

$$\frac{24}{36} = \frac{2}{3}$$

Problem 2 *Follow the steps below to simplify the fraction* $\frac{18}{45}$.

(a) *Find the Greatest Common Factor (GCF) of 18 and 45.*

$$GCF = 9$$

(b) *Divide both the numerator and the denominator by the GCF to simplify.*

$$\frac{18}{45} = \frac{2}{5}$$

Hint: *List the factors for each number to find the largest one they share (GCF):*

- *Factors of 18: 1, 2, 3, 6, 9, 18*
- *Factors of 45: 1, 3, 5, 9, 15, 45*

Procedure 2 (Simplifying Fractions Using Prime Factorizations). (a) Write the prime factorization of the numbers in the numerator and denominator.

(b) Divide out the common factors that appear in both the the numerator and denominator.

(c) Multiply the remaining numbers in the numerator and denominator to find the result.

Remark 1. To **divide out the common factors** means we remove them, since their quotient is 1.

Example 2. Simplify the fraction $\frac{20}{25}$ using prime factorizations.

Explanation. Begin by writing the prime factorizations of 20 in the numerator and 25 in the denominator. Since there is a common factor of 5 in the numerator and denominator, we can divide out the fives since $\frac{5}{5} = 1$. Finally, we multiply the remaining numbers in the numerator and denominator to find the result.

$$\begin{aligned}\frac{20}{25} &= \frac{2 \cdot 2 \cdot 5}{5 \cdot 5} \\ &= \frac{2 \cdot 2 \cdot \cancel{5}}{5 \cdot \cancel{5}} \\ &= \frac{4}{5}\end{aligned}$$

Problem 3 *Simplify the fraction using the prime factorizations shown below.*

$$\begin{aligned}42 &= 2 \cdot 3 \cdot 7 \\ 70 &= 2 \cdot 5 \cdot 7\end{aligned}$$

$\frac{42}{70} = \frac{3}{5}$

Hint: *Write the prime factorizations in the fraction, then cancel the common prime numbers.*

$$\frac{42}{70} = \frac{2 \cdot 3 \cdot 7}{2 \cdot 5 \cdot 7}$$

Problem 4 *Simplify the fraction using whichever method you prefer.*

$$\frac{33}{60} = \frac{11}{20}$$

Hint: *If you want to use prime factorizations, write the prime factorizations in the fraction, then cancel the common prime numbers.*

$$\frac{33}{60} = \frac{3 \cdot 11}{2 \cdot 2 \cdot 3 \cdot 5}$$

Hint: *If you want to use the Greatest Common Factor (GCF) to simplify, the factors of* 33 *are* $1, 3, 11, 33$. *The largest factor that evenly divides into* 60 *is the GCF.*

Problem 5 *Simplify the fraction using whichever method you prefer.*

$$\frac{60}{126} = \frac{10}{21}$$

Hint: *If you want to use prime factorizations, write the prime factorizations in the fraction, then cancel the common prime numbers.*

$$\frac{60}{126} = \frac{2 \cdot 2 \cdot 3 \cdot 5}{2 \cdot 3 \cdot 3 \cdot 7}$$

Hint: *If you want to use the Greatest Common Factor (GCF) to simplify, the factors of* 60 *are* $1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60$. *The largest factor that evenly divides into* 126 *is the GCF.*

Problem 6 *A recycling center collected* 54 *tons of plastic last month. However, due to contamination (like food waste or non-recyclable items), only* 21 *tons could be recycled. What* ***fraction*** *of the collected plastic was recycled?* ***Remember to simplify your answer, if possible****.*

$$\textit{Fraction Recycled} = \frac{7}{18}$$

Hint: *Write the fraction as the amount recycled over the total amount collected, then simplify.*

$$\frac{21}{54}$$

Problem 7 *An animal rescue organization saved* 120 *dogs from an overcrowded shelter. Of these dogs,* 45 *were seniors who required specialized medical care and longer stays before they were ready for adoption. What* ***fraction*** *of the rescued dogs were seniors?* ***Remember to simplify your answer, if possible****.*

$$\textit{Fraction of Senior Dogs} = \frac{3}{8}$$

Hint: *Write the fraction as the number of senior dogs over the total number of dogs rescued, then simplify.*

$$\frac{45}{120}$$